OCEANENVIRO WINCH INTEGRATION WITH FLIGHT

SETUP AND USER GUIDE

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1 Introduction

This document describes the basic guidelines of how to integrate the EIVA OceanEnviro winch with Flight for PC remote control.

The OceanEnviro winch integration with Flight is used for ScanFish or ViperFish operations. This integration allows the user to have the winch cable counter and speed information displayed in Flight and also to remotely control the winch through the Flight software.

2 Physical setup

The first step is to create a physical network connection between the OceanEnviro winch and the Flight computer. This allows the Flight software to control and receive cable and speed data from the OceanEnviro winch over network

Connect the Flight computer with an Ethernet Cat 6 cable to the Simatic PLC inside the electronic cabinet of the winch. Figure 1 below shows the physical installation inside the electronic cabinet.

Figure 1 LAN cable connection inside the electronic cabinet in the winch

The other end of the Ethernet cable should be plugged directly into one of the Flight computer's network interface cards (NIC) ports or be integrated into the general survey network through a switch.

3 Network setup

Next, configure a network to allow communication between the Flight computer and the winch.

1. To configure the network interface card on the computer that the Ethernet cable is connected to, click **Start > Control Panel**.
2. Click **Network and Sharing Center** in the **Control Panel**.
3. On the left pane, click **Change adapter settings**, right click the allocated NIC and then select **Properties**.
4. On the **Networking** tab, click **Internet Protocol Version 4 (TCP/IPv4)** and then click **Properties**.
5. Select **Use the following IP address** and then enter an IP address with the following values: **192.168.0.xxx** (where xxx is a value between 2-255 that is

different from any other devices on the network on the same scope to avoid IP conflict).

6. Enter the Subnet mask **255.255.255.0**.
7. If this NIC has other scopes configured already, go to **Advanced** and add the IP scope.
8. Click **OK** and then click **Close.**

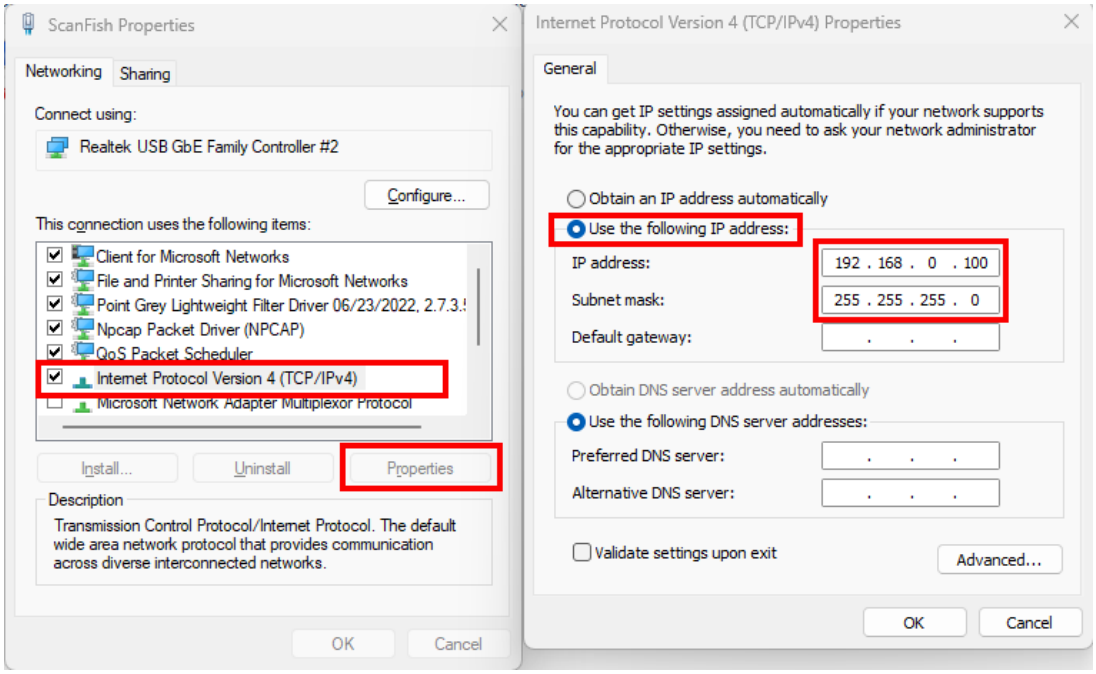

Figure 2 Configuring static IP for the OceanEnviro network scope

4 Flight setup

Open Flight and go to **Settings** > **Options**. Under **Winch and Cable Counter,** connections and configurations for both the winch and cable counter can be set up. The data from the cable counter is displayed automatically in the **Numeric panel**, shown in Figure 3, while data chart(s) for cable counter length and speed may be set up as shown in Figure 4.

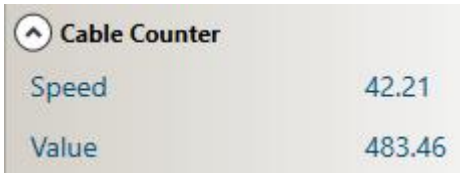

Figure 3 Cable counter values in the Numeric panel

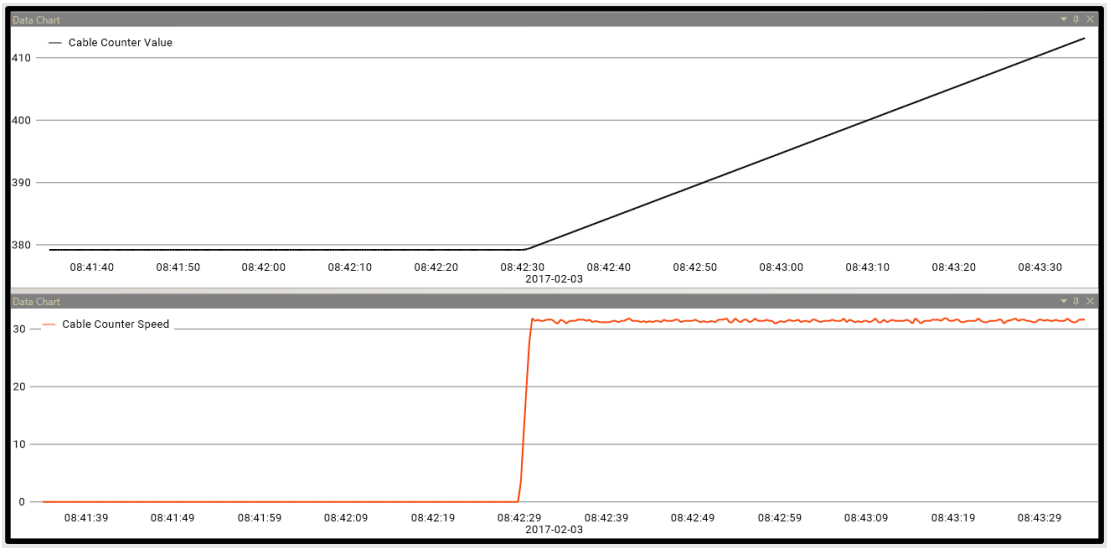

Figure 4 Data charts for cable counter value and speed

When **Winch** is highlighted in the **Options** dialogue box, the properties shown in Figure 5 will appear in the **Properties Pane** on the right side of the **Options** dialogue box.

Type	OceanEnviro
Connection	tcp://192.168.0.1:3000/
Minimum Limit	10 m
Maximum Limit	700 m
Repeat Interval	200 ms
Ignore Safety Limits	✖ False
Allow Auto-Winch	✖ False
Speed Out	10 %
Speed In	10 %

Figure 5 Winch properties

The winch properties are explained below:

- In **Type** the user should select the correct winch driver.
  - **Note**: Flight only supports two types of winches, the EIVA Ocean Enviro and a winch from KC Denmark.
- In **Connection,** the user can set the I/O connection up. Open the I/O Selector by highlighting the **Connection** line and click the three-dotted button to the right. UDP, TCP and serial connection are supported. The default connection and protocol to EIVA OceanEnviro winches are TCP: 192.168.0.1 port: *any*. The I/O Selector has a test function and is identical to the I/O selector below.

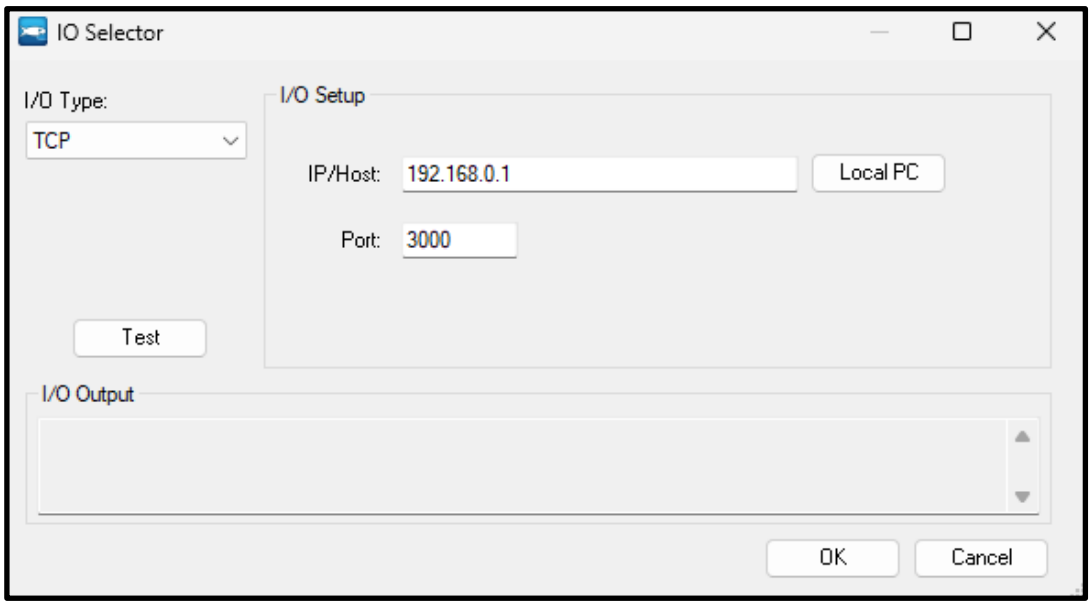

Figure 6 I/O Selector

- **Minimum limit** is a safety measure that only allows winch control if the cable out value from the cable counter is greater than the value specified here. This prevents the pilot from pulling the ROTV through the cable sheave.
    - **Note**: this only applies when **Ignore safety limits** is set to **False**.
- **Maximum limit** is a safety measure that only allows winch control if cable out value from the cable counter is less than the value specified here. This prevents the pilot from winching more cable out than the **Maximum limit**.
    - **Note**: This only applies when **Ignore safety limits** is set to **False**.
- **Repeat interval** is a value in milliseconds between signal repetitions (in/out). Winches have different requirements here: for the KC Denmark winch the value should be greater than 700 ms, while the EIVA Ocean Enviro should be less than 500 ms.
- **Ignore safety limits** makes it possible to ignore the settings set in **Minimum limit** and **Maximum limit**.
- **Allow Auto-Winch** is a function to allow automatic cable length adjustments to the depth of the ROTV. There are different types of automated winch drives.

When **Cable Counter** is highlighted in the **Options** dialogue box, the properties shown in Figure 7 will appear in the **Properties** pane on the right side of the **Options** dialogue box.

Figure 7 Cable counter properties

The properties are described below:

- **Integrated in winch** is an option to use the same I/O setup as the **Winch Connection** where applicable. If set to **True,** all other properties are greyed out.
- If the cable counter is not integrated in the winch control, or winch control is not used, the user can freely choose between different drivers (Figure 8) for the cable counter values.
- In **Connection** the user can set the I/O connection up. Open the **I/O Selector** by highlighting the **Connection line** and click the three-dotted button to the right. UDP, TCP and serial connections are supported.

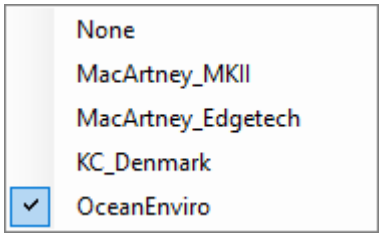

Figure 8 Cable Counter drivers

Note: It is possible to use and test the OceanEnviro winch without any ROTV (ScanFish/ViperFish) connected in Flight. To do this, go to **Settings** > **Options** > **Fish** > **Driver** and set to **None**. If not done, Flight will complain about missing connection to the driver selected when clicking **Connect**.

5 Flight winch buttons

In the Flight toolbar, the winch control buttons are listed, from left to right, as **Cable Auto** mode, **Cable In**, **Cable Stop** and **Cable Out**. The user will need to hold the left mouse button down to winch in or out on the respective buttons. When winch is active in any of the modes, the mark in the upper right corner is green. If the auto-winch is not configured, like in Figure 9 below, it is greyed out.

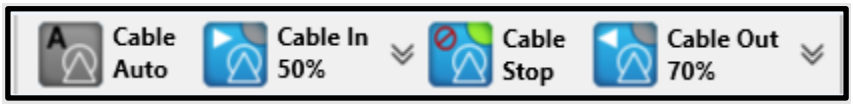

Figure 9 Winch control buttons in toolbar

The speed of the winch can be controlled from the dropdown windows accessible for both **Cable In** and **Cable Out** buttons in the Flight toolbar. It is in 10%-step intervals of the total max speed. The total max speed depends on the model and the layer on the drum. The selected speed is listed on the button itself; 50% under **Cable In** in Figure 10.

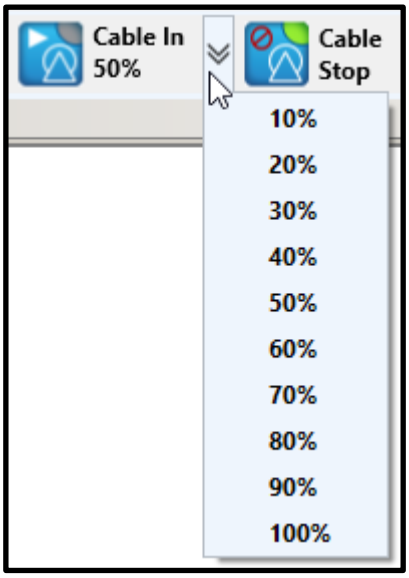

Figure 10 Winch speed control drop-down window